# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: INTERNET PROGRAMMING**
**COURSE CODE: CSA43**

**COURSE OUTCOME COVERED**

**CO-2 :** Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Integrated Experiments

Weightage: 40%

**QUESTIONS: Total Marks: 30**

## Experiment 1: Responsive Layout Design

Suppose that a retail website is not responsive across mobile and tablet devices, causing misalignment of content and improper spacing.

**(a) Analyze the possible reasons for lack of responsiveness in the existing layout.**

A responsive website automatically adjusts its layout according to the screen size of the device. If a retail website is not responsive, users may experience overlapping text, improper spacing, horizontal scrolling, and poor readability.

**Possible Reasons:**

1. Fixed Width Layout
    - The webpage uses fixed widths such as `width:1200px` instead of flexible widths (`100%`, `auto`, `%`).
    - Fixed layouts cannot adapt to different screen sizes.
2. No Viewport Meta Tag
3. Without the viewport tag, mobile browsers display the webpage as a desktop.

**(b) Design a responsive webpage layout using CSS Flexbox or Grid, ensuring proper alignment of header, navigation, content, and footer.**

```html
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Responsive Layout</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<header>
<h1>Online Shopping</h1>
</header>

<nav>
<a href="#">Home</a>
<a href="#">Products</a>
<a href="#">Offers</a>
<a href="#">Contact</a>
</nav>

<div class="container">

<section class="content">
<h2>Welcome</h2>
<p>This is the main content area.</p>
</section>

<aside class="sidebar">
<h3>Categories</h3>
<ul>
<li>Electronics</li>
<li>Fashion</li>
<li>Books</li>
</ul>
</aside>

</div>

<footer>
```

```
Copyright © 2026
</footer>

</body>
</html>
```

**(c) Demonstrate how the layout adapts to different screen sizes (mobile, tablet, desktop) using media queries.**

```
@media screen and (max-width:768px){

.container{
flex-direction:column;
}

nav{
flex-direction:column;
}

nav a{
text-align:center;
border-bottom:1px solid white;
}

}
```

**(d) Implement spacing, alignment, and distribution techniques to improve visual consistency.**

```
.container{
display:flex;
justify-content:space-between;
align-items:flex-start;
gap:20px;
padding:20px;
}

.content,
.sidebar{
border-radius:10px;
box-shadow:0 0 10px gray;
padding:20px;
}
```

(e) Evaluate the effectiveness of the responsive design and suggest further improvements.

**Evaluation**

The responsive webpage works efficiently on desktop, tablet, and mobile devices.

**Advantages**

- Supports multiple screen sizes.
- Eliminates horizontal scrolling.
- Proper alignment of all sections.
- Improves readability.
- Enhances user experience.
- Easy to maintain using Flexbox.

## Experiment 2: Form Validation using JavaScript

Suppose that a registration form accepts invalid email and phone number inputs, leading to incorrect data submission.

**(a) Analyze the issues in the existing form validation mechanism.**

Invalid email accepted.
Phone number accepts letters.
Empty fields submitted.
No error messages.
Poor user experience.

**(b) Design input fields for email and phone number with appropriate HTML attributes.**

```
<form>

Email:
<input type="email" id="email" required>

Phone:
<input type="text" id="phone" required maxlength="10">

<p id="msg"></p>

<button onclick="return validate()">Submit</button>

</form>
```

**(c) Implement JavaScript validation using Regular Expressions for both email and phone number.**

```
function validate(){

var email=document.getElementById("email").value;
var phone=document.getElementById("phone").value;

var emailPattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;
var phonePattern=/^[0-9]{10}$/;

if(!emailPattern.test(email)){
alert("Invalid Email");
return false;
}

if(!phonePattern.test(phone)){
alert("Invalid Phone Number");
return false;
}

alert("Validation Successful");
return true;

}
```

**(d) Develop a mechanism to display real-time error messages for invalid inputs.**

```
document.getElementById("email").onkeyup=function(){

let email=this.value;
let pattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;

document.getElementById("msg").innerHTML=
pattern.test(email)?"Valid Email":"Invalid Email";

}
```

**(e) Evaluate the validation system and suggest enhancements for better user experience and security.**

Advantages

- Prevents invalid input.
- Better user experience.
- Reduces server errors.

Enhancements

- Password validation.
- CAPTCHA.
- Server-side validation.

- Input sanitization.

## Experiment 3: CSS Layout Debugging

Suppose that a webpage layout overlaps when the browser window is resized, affecting usability.
**(a) Analyze the CSS properties responsible for layout overlapping issues.**

### Analyze overlapping issues

- Fixed width elements.
- Absolute positioning.
- Missing Flexbox/Grid.
- Overflow hidden.
- Incorrect z-index.

**(b) Examine the role of the CSS box model (margin, border, padding, content) in layout design.**

### CSS Box Model

The box model contains:

- Content
- Padding
- Border
- Margin

Proper use prevents layout overlap.

**(c) Debug the layout using appropriate positioning techniques (relative, absolute, flex, grid).**

```
.container{
display:flex;
flex-wrap:wrap;
gap:20px;
}

.box{
flex:1;
min-width:250px;
padding:20px;
}
```

**(d) Modify the CSS to ensure proper alignment and prevent overlapping during resizing**.

```
*{
box-sizing:border-box;
}

.container{
display:flex;
flex-wrap:wrap;
justify-content:center;
}

img{
max-width:100%;
height:auto;
}
```

**(e) Evaluate the corrected layout and suggest best practices for maintaining consistency.**

**Result**

- No overlapping.
- Responsive layout.
- Better readability.
- Improved usability.

**Best Practices**

- Use Flexbox/Grid.
- Avoid fixed widths.
- Use media queries.
- Use `box-sizing:border-box`.
- Test on different screen sizes.
- Optimize images.

***Code of Conduct:***

*I ___Ch. Bhargav Ram (1922425334)________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***